

A TASK IS USABLE ONLY WHEN IT GETS DONE

A visitor without a smartphone can speak to staff and leave with paper. One missing two-key receipt keeps the affected service closed.



SAME CAMERA · ONLY THE OPERATING MODE CHANGES

0 / 7 SIGNED SEVEN RECEIPTS. ONE MISSING, STAY CLOSED. Two functional signatures. One person cannot do both.	G01 SITE + TIME site controller × duty lead no site release: open nothing KEY 01 + KEY 02 MISSING → CLOSED	G02 ACCESS + SAFETY safety owner × access replayer route changes: room and service stop only KEY 01 + KEY 02 MISSING → CLOSED	G03 SAME TASK ON PAPER service owner × paper operator failed replay: verified manual route only KEY 01 + KEY 02 MISSING → CLOSED	G04 INFO, DATA + EXIT data owner × service owner source or rule changes: assistance off KEY 01 + KEY 02 MISSING → CLOSED	G05 SHIFT TAKEOVER operations owner × named substitute no takeover: manual and assisted closed KEY 01 + KEY 02 MISSING → CLOSED	G06 QUOTE + PURCHASE authorized buyer × acceptance owner no comparison: no order or total KEY 01 + KEY 02 MISSING → CLOSED	G07 OPEN + CLOSE duty lead × independent replayer shift ends: receipts do not carry KEY 01 + KEY 02 MISSING → CLOSED

THREE WORKS, THREE REAL JOBS

Lab to Street controls release. Problem to Change controls maintenance, suspension, and retirement. Technology must be able to return when it fails.



Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · NS89FZ · CC BY-SA 4.0 · External context reference; not field evidence



Student reading room at the University of Science and Technology Beijing; shared-learning and night-study reference
PHOTO-10 · Cyber9 · Public domain dedication by the copyright holder · External context reference; not field evidence



North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
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PROTOTYPE WORKS: CONTAIN FAILURE FIRST

An observation foyer, accessible test lane, isolated test pocket, physical stop, and failure review make a controlled section. A robot that fails here never reaches the public street.

LEARNING WORKS: MAKE THE NEXT STEP CLEAR

Public arrival, Ask First, a shared table, paper material, and human sign-off form a service-and-learning section. A visitor without a smartphone still knows whom to ask.

EVERYDAY WORKS: SURVIVE AN ORDINARY EVENING

A shaded entry, quiet rest, human takeover, and public change record form an everyday section. Dazhongsi siting remains unverified, so every component stays movable.

COMPONENT
C04 accessible test lane

OWNER
Test lead and site safety

RESOURCE
M-class movable kit

COMPONENT
C01 Ask First + C05 marker

OWNER
Service desk and venue

RESOURCE
S-class map, signs, staff

COMPONENT
C02 Takeover + C03 Change Log

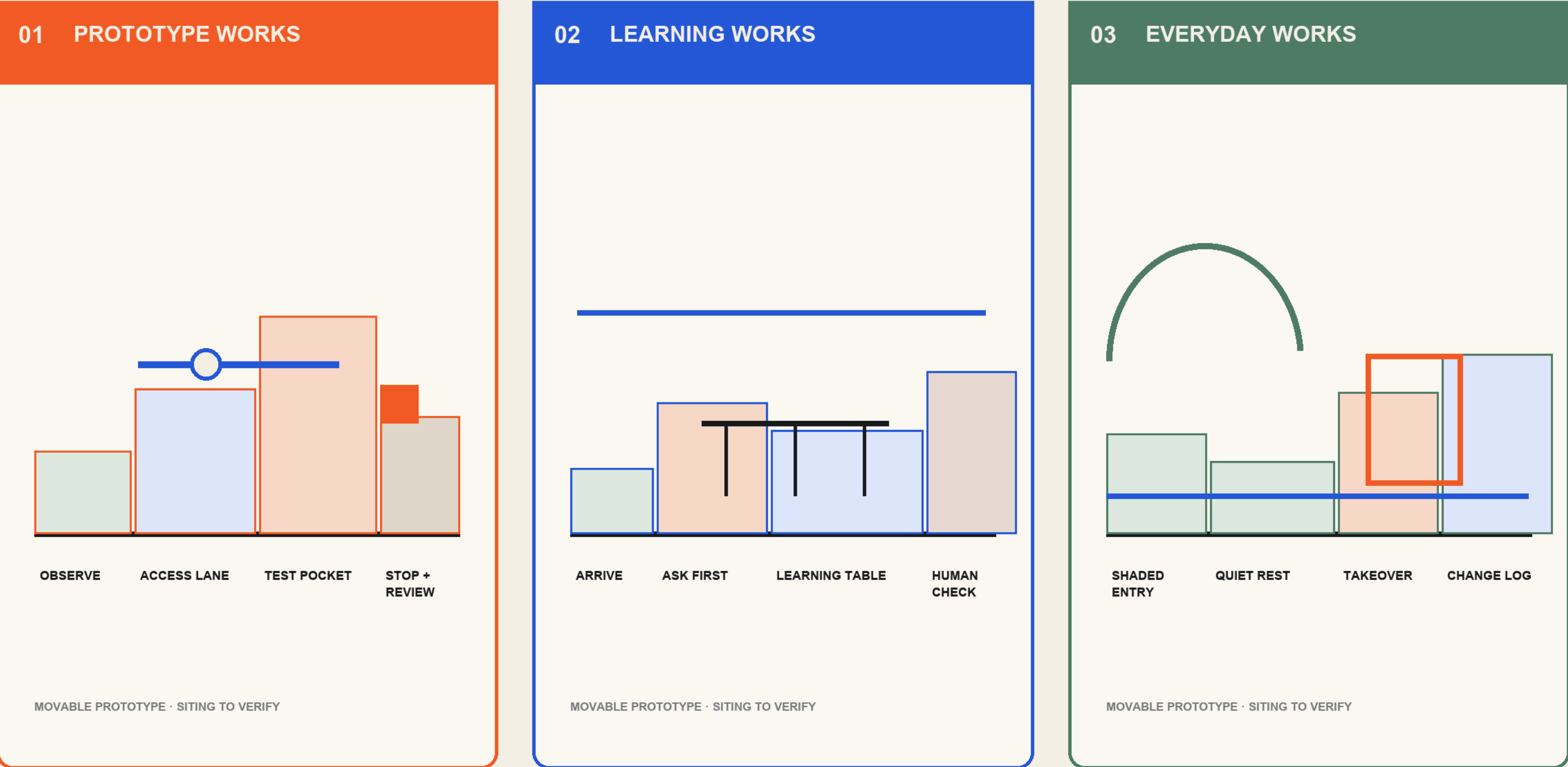
OWNER
Safety lead and service owner

RESOURCE
S/M movable kit

JING-ZHANG WORKS

THREE WORKS, THREE SPATIAL PROTOTYPES

One service rule becomes controlled testing, public learning, and everyday service space



PROVISIONAL GEOMETRY · EPSG:4326 DISPLAY / EPSG:4548 AREA CHECK

01—05

WRITE DOWN HOW ONE SHIFT OPENS

Reference workload is inspectable. People, roster, quotations, budget, permits, and field execution await authorization.

WRITE THE CLOCK, WORKLOAD, AND STOP RULE BEFORE OPENING

EIGHT REFERENCE PUBLIC HOURS · 21.5 FUNCTIONAL-ROLE HOURS · SEVEN CHECKPOINTS · SIX QUOTE CARDS · ZERO SUPPLIER QUOTATIONS

SEVEN CHECKPOINTS THAT MUST LEAVE EVIDENCE

T-60	T-30	T-10	T+00	T+120	T+240	T+480
walk full route	replay on paper	sign release	show state	independent replay	run fault card	close + reconcile

FIVE FUNCTIONAL WORKLOADS

duty lead	9.75h	route through close
paper service	9h	paper task + takeover
access replay	1h	pre-open + midday
safety fault	1h	stop + fault card
information + data	0.75h	source + deletion
21.5 h functional coverage, not people, roster, wages, or legal working time		

SIX EVENT DEADLINES

01	before next public transaction	safety / privacy / critical information / network
02	before paper is issued	staff accepts human takeover
03	before visitor leaves	give issue ID + review date
04	before closing sign-off	reconcile issues, versions, deletion
05	before public reopening	recovery evidence + independent replay
06	after supplier exit	disconnect, export allowed records, restore

SIX QUOTE-READY WORK CARDS

C01 6 routes + 12 observations	C02 3 paper sets + 1 master	C03 one 6.0 × 4.8m set	C04 one test at a time	C05 21.5 functional hours	C06 permanent works zero
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REFERENCE WORK ORDER · 0 SUPPLIER QUOTATIONS · TOTAL NULL · SITE, DATE, PEOPLE, SHIFT COUNT, PERMITS, AND FIELD RESULTS EMPTY

21.5 functional-role hours compare workload only. Supplier quotations are zero, monetary total is null, and field results are zero.